

Research Methodology in CS (CSEG3060_3)

Lecture 4: Patenting Process - Development and Filing: Patentability criteria, filing procedures (provisional and complete specification), examination, publication, opposition, and grant of patents; commercialization and development aspects.

Unit 5: Intellectual Property and Patents • Mapped Course Outcome: CO2

Teaching Notes: Research Methodology in Computer Science

Course: CSEG3060_3 | Program: B.Tech (CS) | Semester 5

Unit 5: Intellectual Property and Patents

Lecture 4: Patenting Process - Development and Filing

Patentability Criteria, Filing Procedures, Examination, Publication, Opposition, Grant, and Commercialization

1. Lecture Overview and Learning Objectives

1.1 Lecture Context

This lecture provides an in-depth examination of the complete patenting lifecycle, beginning from the conceptualization of an invention through to its grant and subsequent commercialization. It builds upon Lecture 3 of this unit, which addressed the nature of intellectual property and the broader patent landscape. Here, the focus narrows specifically to the procedural and substantive dimensions of patenting, equipping students with a practical understanding of how an idea transforms into a legally protected, marketable asset.

For computer science researchers, this knowledge is critical because the procedural rigor of patenting is as consequential as the technical novelty of the invention itself. A technically groundbreaking invention may fail to secure patent protection if procedural lapses occur, while a relatively modest innovation may receive robust protection if correctly documented and prosecuted.

1.2 Learning Objectives

By the end of this lecture, students should be able to:

1. **Identify** and **apply** the substantive criteria of patentability (novelty, inventive step, industrial application) to a given invention, particularly in software and computing domains.
2. **Differentiate** between provisional and complete specifications and determine the appropriate stage to file each.
3. **Describe** the procedural steps from patent filing to grant, including examination, publication, and opposition.
4. **Evaluate** the grounds for patent opposition under the Indian Patents Act, 1970 (as amended in 2005) and international equivalents.
5. **Analyze** strategies for commercialization of patented inventions, including licensing, assignment, and technology transfer.
6. **Recognize** the role of the Patent Cooperation Treaty (PCT) in international filing strategy.

1.3 Mapped Course Outcome

CO2: Develop the ability to understand and apply intellectual property regimes, including the procedural and substantive requirements of patenting, to protect research outputs in computer science.

2. Patentability Criteria: The Substantive Foundation

Before any procedural step is initiated, an invention must satisfy the substantive criteria of patentability. These are universal principles recognized across major patent jurisdictions, though their specific interpretation may vary slightly between national laws (e.g., the Indian Patents Act, 1970, the European Patent Convention, and the United States Patent Act, 35 U.S.C.).

2.1 The Three Pillars of Patentability

2.1.1 Novelty

Novelty requires that the invention must be new, meaning it should not have been anticipated by any prior art available anywhere in the world before the date of filing (or the priority date). Under Section 2(1)(j) of the Indian Patents Act, "novelty" is defined in conjunction with "new invention," which means any invention or technology that has not been anticipated by publication in any document or used in the country or elsewhere in the world before the date of filing.

Key principles:

- **Absolute Novelty:** Indian law adopts the standard of universal novelty, encompassing disclosures anywhere in the world.
- **Prior Art Definition:** Prior art includes any published matter (patent, journal article, conference paper, blog post, open-source code release) or public use or knowledge that discloses the same invention.
- **Grace Period:** Under Section 31 (now superseded by Section 11A and related provisions), and currently under the Patents (Amendment) Act 2005, India provides a 12-month grace period for disclosures arising from exhibitions or academic publications by the inventor, allowing protection despite prior publication.

For computer science, novelty questions frequently arise around:

- Open-source software releases that disclose algorithmic innovations
- Conference publications (e.g., NeurIPS, SIGCOMM) that predate patent filings
- Public demonstrations at hackathons or product launches

2.1.2 Inventive Step (Non-Obviousness)

Section 2(1)(ja) of the Indian Patents Act defines "inventive step" as a feature of an invention that involves a technical advance as compared to the existing knowledge or having economic significance or both, and that makes the invention not obvious to a person skilled in the art.

The "person skilled in the art" is a legal fiction representing an ordinary practitioner in the relevant field with average knowledge and skill. The test is whether, given the prior art, such a person would have found the invention obvious.

In the United States (35 U.S.C. § 103), the analogous standard is "non-obviousness," assessed against the *Graham v. John Deere Co.* (1966) framework, which considers:

1. Scope and content of the prior art
2. Differences between the prior art and the claims
3. Level of ordinary skill in the pertinent art
4. Secondary considerations (commercial success, long-felt need, failure of others)

Example: An algorithm that merely combines known sorting techniques without producing a measurable technical advance would lack an inventive step, even if the combination is novel.

2.1.3 Industrial Application (Utility)

Section 2(1)(ac) defines "capable of industrial application" in relation to an invention, meaning that the invention is capable of being made or used in an industry. The term "industry" is broadly

interpreted under Section 2(1)(m) to include any manufacturing, service, or commercial activity.

For software patents, this criterion requires that the invention must have a specific, substantial, and credible utility, going beyond abstract ideas. This is particularly significant in jurisdictions such as the United States, where the Supreme Court's decisions in *Alice Corp. v. CLS Bank International* (2014) established a two-step test:

Step 1: Determine whether the claims at issue are directed to a patent-ineligible abstract idea (e.g., algorithms, methods of organizing human activity, mental processes).

Step 2: If so, examine whether the additional elements transform the nature of the claim into a patent-eligible application, i.e., an "inventive concept."

2.2 Statutory Exclusions from Patentability

Section 3 of the Indian Patents Act enumerates categories that are not inventions and therefore not patentable. For computer science researchers, the following are particularly relevant:

Exclusion	Description	CS Relevance
Section 3(a)	Frivolous inventions contrary to natural laws	Pure mathematical methods without specific application
Section 3(k)	A mathematical method or business method or a computer program per se or algorithms	Software per se, algorithms without technical application
Section 3(m)	A mere arrangement or re-arrangement or duplication of known devices	Routine UI changes without technical effect
Section 3(h)	Methods of agriculture or horticulture	Computational methods for agriculture may still be patentable if they constitute a technical solution

Important Clarification on Section 3(k): The Indian Patent Office's revised guidelines (2017 and subsequent amendments) acknowledge that while computer programs "per se" are excluded, inventions involving software that produce a "technical effect" or contribute to a "technical advancement" are patentable. For example, a software-implemented encryption method that improves security architecture may be patentable, but a business method implemented entirely in software for accounting purposes may not be.

3. Patenting Process: Pre-Filing Considerations

3.1 Patent Searching

Before filing, a thorough prior art search is essential. Tools include:

- **Patent databases:** USPTO (PatFT, AppFT), EPO (Espacenet), Indian Patent Advanced Search System (IPAIRS), Google Patents, WIPO PATENTSCOPE.
- **Non-patent literature:** IEEE Xplore, ACM Digital Library, arXiv, Springer, ScienceDirect.

A well-conducted search helps in:

- Assessing novelty and inventive step
- Refining claim scope to avoid infringement
- Identifying potential freedom-to-operate issues
- Avoiding wasted filing costs

3.2 Documentation and Record-Keeping

Inventors should maintain a contemporaneous lab notebook or digital research journal with:

- Date and inventor of each conception
- Detailed description of how the invention works
- Alternative embodiments explored
- Witness signatures (preferably from non-inventors)

In research environments, this documentation serves both as evidence of invention date (in interferences) and as the substrate from which the patent specification is drafted.

4. Filing Procedures: Provisional and Complete Specifications

4.1 Conceptual Distinction

The Indian Patents Act (Sections 7-11) provides two distinct stages for filing:

- **Provisional Specification:** A preliminary document establishing the priority date and disclosing the broad contours of the invention.
- **Complete Specification:** A definitive document containing the full technical disclosure and the legal claims defining the scope of protection sought.

4.2 Provisional Specification

4.2.1 Purpose

The provisional specification allows the inventor to secure an early priority date, typically when:

- The invention is at a conceptual or experimental stage
- Further research and refinement are anticipated
- Market or publication pressures require an early filing date
- Investor or funding discussions require IP position establishment

4.2.2 Contents (Rule 15)

A provisional specification must contain:

1. Title of the invention
2. Description of the invention (brief and indicating its nature, purpose, and components)
3. Any drawings referenced

It does **not** require claims, abstract, or detailed embodiments.

4.2.3 Limitations

- **12-Month Deadline:** The complete specification must be filed within 12 months of the provisional filing date (Section 9(1)). Failure to do so results in the provisional application being deemed abandoned, and the priority date is lost.
- **Disclosure Sufficiency:** The provisional must enable a person skilled in the art to understand the invention broadly. Later-filed complete specifications can include additional matter but cannot claim priority for matter not disclosed in the provisional.

4.3 Complete Specification

4.3.1 Purpose

The complete specification is the substantive patent document, providing full disclosure of the invention and the legal metes and bounds of the protection sought.

4.3.2 Contents (Rule 13)

A complete specification typically comprises:

1. **Title of the Invention:** Concise, descriptive, and aligned with classification systems.
2. **Field of Invention:** A statement indicating the technical area.
3. **Background and Prior Art:** Description of existing solutions and their limitations.
4. **Objects of the Invention:** The technical problems the invention addresses.
5. **Summary of the Invention:** High-level overview of the solution.
6. **Brief Description of Drawings:** Where applicable.
7. **Detailed Description:** A full, enabling disclosure sufficient for a person skilled in the art to perform the invention. Must satisfy the enablement requirement (Section 10(4)).
8. **Claims:** The most legally significant section, defining the scope of the monopoly. Claims must be clear, concise, and fully supported by the description.
9. **Abstract:** A brief technical summary (up to 150 words) for publication purposes.

4.3.3 Types of Claims

- **Independent Claims:** Stand alone and define the broadest protection.
- **Dependent Claims:** Reference one or more earlier claims and add further limitations, providing fallback positions during prosecution.
- **Product Claims:** Cover apparatus, system, device, or composition.
- **Process Claims:** Cover methods or procedures.
- **Use Claims:** Cover specific applications of known products.

For software inventions, typical claim structures may include:

- A system claim (e.g., "A distributed computing system comprising...")
- A method claim (e.g., "A computer-implemented method comprising...")
- A computer-readable medium claim (e.g., "A non-transitory computer-readable storage medium storing instructions that, when executed, cause...")

4.4 Filing Modes

Mode	Description	Typical Use
Physical Filing	Submission at appropriate Patent Office (Delhi, Mumbai, Chennai, Kolkata)	Traditional, established workflow
e-Filing	Online filing via ipindiaonline.gov.in	Faster processing, instant acknowledgment
PCT National Phase	Filing in India under PCT within 31 months of priority date	International applicants
Paris Convention	Foreign applicants claiming priority from home country filing within 12 months	Convention-based filings

5. Examination Procedure

5.1 Request for Examination (Section 11B)

Unlike many jurisdictions where examination is automatic, India follows an **examination-on-request** system. The applicant (or any interested person) must file a Request for Examination (Form 18) within **48 months** of the priority date or filing date, whichever is earlier.

Failure to file within this period results in the application being treated as withdrawn.

5.2 First Examination Report (FER)

Upon receipt of the examination request, the application is assigned to an examiner who conducts a substantive examination against the patentability criteria. The examiner issues a First Examination Report (FER) citing:

- Prior art references (typically classified as X, Y, A categories)
- Deficiencies in form or substance
- Objections regarding novelty, inventive step, or industrial application
- Clarity, succinctness, or support objections under Section 10

The applicant must respond to the FER within **six months** (extendable by a further three months on a request in Form 4) from the date of issuance.

5.3 Prosecution and Hearing

The prosecution involves an iterative exchange:

1. Applicant files response addressing each objection with arguments and/or amendments.
2. Examiner reviews and may issue further objections (Examination Report).
3. If objections persist, a hearing may be requested under Section 14, where the applicant may appear before the controller to argue the case orally.
4. The Controller may allow or refuse the application.

The final outcome is either:

- **Allowance of claims** (with or without amendments)
- **Refusal of application**

A refused application may be appealed to the Intellectual Property Appellate Board (IPAB), which has been succeeded by the Commercial Courts and Intellectual Property Divisions of High Courts under the Tribunals Reforms Act, 2021.

6. Publication of Patent Application

6.1 Statutory Basis and Timeline

Under Section 11A of the Indian Patents Act, a patent application is published in the Patent Journal **18 months** after the priority date or filing date. Publication may occur earlier if the applicant files an Early Publication Request (Form 9).

6.2 Effects of Publication

Publication confers several significant effects:

1. **Constructive Notice:** The public is deemed to have notice of the application.
2. **Provisional Rights:** Under Section 11A(7), the applicant acquires the right to claim damages for infringement occurring after publication, provided that:
 - The patent is ultimately granted.
 - The infringing act would have infringed the granted patent.
 - The applicant gives notice to the infringer before the institution of infringement suits.
3. **Examination Trigger:** Publication often prompts examination requests by third parties who may have commercial interests.

4. Pre-Grant Opposition Period: Publication opens the 6-month window for third parties to file pre-grant oppositions under Section 25(1).

6.3 Confidentiality Considerations

Prior to publication, applications are kept confidential. Inventors should be cautious about:

- Disclosing the invention to unauthorized parties
- Publishing academic papers that may constitute prior art against themselves (grace period applies)
- Discussing the invention at public forums without non-disclosure agreements

7. Opposition Proceedings

7.1 Pre-Grant Opposition (Section 25(1))

Any person may file a pre-grant opposition at any time after publication and before the grant of the patent. Grounds for opposition include:

Ground	Section	Description
Non-patentability under Section 3	Section 25(1)(d)	Invention falls within excluded categories
Lack of novelty	Section 25(1)(d)(i)	Anticipation by prior publication or use
Lack of inventive step	Section 25(1)(d)(ii)	Obvious to person skilled in the art
Insufficient disclosure	Section 25(1)(g)	Specification does not sufficiently describe the invention
False suggestion or wrong mention	Section 25(1)(h)	Misrepresentation in the specification
Non-compliance with Section 8	Section 25(1)(j)	Failure to disclose corresponding foreign applications

Representation is by way of a notice of opposition (Form 7A) supported by evidence and arguments.

7.2 Post-Grant Opposition (Section 25(2))

Following grant, any interested person may file a post-grant opposition within **12 months** of the date of grant notification, or the publication of the grant in the journal, whichever is later.

Grounds largely mirror pre-grant opposition but additionally include:

- Non-disclosure of information required under Section 8
- Fraud on the patent office
- Claiming of wrong priority date

7.3 Strategic Implications

Opposition is often used by competitors as a defensive strategy to:

- Block or limit patent monopolies in critical technology areas
- Delay or invalidate patents that threaten market position
- Test the validity of weak or overly broad patents

For patent applicants, the possibility of opposition necessitates careful drafting, thorough prior art consideration, and robust responses to any pre-grant oppositions.

8. Grant of Patent

8.1 Grant Procedure

Upon successful prosecution and resolution of all objections, the Controller grants the patent under Section 43. The grant is recorded in the Register of Patents, and the patent number is issued.

8.2 Patent Certificate

The granted patent includes:

- Patent number and date of grant
- Priority date and filing date
- Name of patentee and inventor
- Full specification as granted
- Term of the patent

8.3 Term of Patent

Under Section 53, the term of a patent in India is **20 years** from the date of filing (whether provisional or complete), subject to payment of renewal fees.

Renewal fees (under Rule 80) are payable annually from the third year onwards. Failure to pay results in lapse, though restoration is possible within 18 months of lapse with a prescribed surcharge.

8.4 Rights Conferred by Grant (Section 48)

A granted patent confers on the patentee the exclusive right:

- To prevent third parties from making, using, offering for sale, selling, or importing the patented product or process
 - To assign or license the patent
 - To seek reliefs for infringement, including damages, injunctions, and accounts of profits
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9. Commercialization and Development Aspects

9.1 The Commercialization Imperative

A patent is not merely a legal right; it is a strategic commercial asset. The value of a patent is realized only when it is actively commercialized or otherwise deployed. For computer science patents, commercialization pathways include:

9.2 Commercialization Strategies

9.2.1 Direct Commercialization by the Patentee

The patent owner manufactures, markets, and sells the patented product or services themselves. This is most common when:

- The patentee is a company with manufacturing or service delivery capabilities
- The invention is core to the patentee's business model

9.2.2 Licensing

Licensing allows the patentee to retain ownership while granting defined rights to a licensee in exchange for consideration. Common licensing structures include:

- **Exclusive License:** Sole licensee with rights excluding even the patentee.
- **Sole License:** Only one licensee but the patentee may also practice.
- **Non-Exclusive License:** Multiple licensees permitted.
- **Cross-Licensing:** Mutual licensing arrangements, common in standards-essential patents.

Licensing terms typically include:

- Upfront fees
- Running royalties (percentage of revenue)
- Minimum royalties
- Milestone payments
- Field-of-use restrictions
- Geographic limitations

9.2.3 Assignment (Sale)

A complete transfer of ownership, typically a one-time payment, structured via an Assignment Deed under Section 69 of the Patents Act. Assignments must be registered with the Controller to be effective against third parties.

9.2.4 Spin-Off Companies

Academic institutions increasingly support spin-off companies based on patented technologies developed in research laboratories. This model preserves academic ownership while enabling commercial exploitation through dedicated entrepreneurial ventures.

9.3 Technology Transfer

Technology transfer is the broader process by which patented inventions transition from research environments to commercial deployment. Mechanisms include:

- University-industry collaborative research agreements
- Government-funded research translation programs
- Patent pools and open innovation platforms
- Standards-setting organization (SSO) patent commitments

9.4 Valuation of Patents

Patent valuation is critical for transactions, licensing negotiations, accounting, and litigation.

Methods include:

- **Cost Method:** Sum of R&D and patent procurement costs
- **Market Method:** Comparable transaction analysis
- **Income Method:** Discounted cash flow analysis of future expected income
- **Real Options Method:** For patents with uncertain but potentially high-value applications

In software-related patents, valuation often hinges on the patent's role in:

- Defensive patent portfolios (reducing litigation risk)
- Offensive litigation positions
- Cross-licensing leverage in standards-essential technology areas (e.g., 5G, Wi-Fi, video codecs)

9.5 Strategic Considerations for Software Patenting

Computer science patents require nuanced commercialization strategy due to:

1. **Rapid Technological Evolution:** Software patents may become obsolete quickly due to evolving technologies.
 2. **Defensive Aggregation:** Companies often build large defensive portfolios to deter litigation (e.g., the LOT Network for open-source-friendly companies).
 3. **Open Source Compatibility:** Many software patents are licensed under FRAND (Fair, Reasonable, and Non-Discriminatory) or open-source-friendly terms to facilitate ecosystem participation.
 4. **Standards Integration:** Patents incorporated into technical standards (e.g., H.265, 5G NR) carry significant licensing obligations.
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10. International Patenting: PCT and Beyond

10.1 Patent Cooperation Treaty (PCT)

The PCT, administered by WIPO, provides a unified procedure for filing patent applications in multiple jurisdictions.

10.1.1 PCT Process

1. **International Phase:**
 - Filing of PCT application (within 12 months of priority filing)
 - International Search Report (ISR) and Written Opinion by International Searching Authority

- Optional International Preliminary Examination (now "International Preliminary Report on Patentability")
- Publication 18 months from priority date

2. National Phase:

- Within 30 or 31 months from priority date (depending on jurisdiction), applicants enter national/regional phases in desired patent offices
- Each office conducts its own substantive examination under domestic law

10.1.2 Strategic Advantages of PCT

- **Deferral of Costs:** National filing costs deferred by 18 months
- **Extended Search Report:** Useful assessment of patentability before commitment
- **International Recognition:** PCT application establishes a tentative patent application in all PCT member states
- **Examination Support:** ISR and Written Opinion inform prosecution strategies

10.2 Paris Convention Priority

For applicants not utilizing PCT, the Paris Convention provides for a 12-month priority period within which to file in other member countries while retaining the original priority date.

10.3 Regional Patent Systems

- **European Patent Convention (EPC):** Single patent grant covering multiple European states via the European Patent Office.
- **ARIPO (African Regional Intellectual Property Organization):** Regional patent system for African states.
- **OAPI (Organisation Africaine de la Propriété Intellectuelle):** Another regional system for francophone African states.
- **Eurasian Patent Convention:** Regional coverage of Eurasian states.

11. Case Study: A Software Patent Prosecution Journey

Consider a hypothetical case involving a novel cryptographic protocol implemented in blockchain systems for secure multi-party computation:

Step 1: University researchers develop a new zero-knowledge proof system that reduces verification time by 40 percent.

Step 2: Researchers file a provisional specification within India outlining the protocol, performance metrics, and broad application areas.

Step 3: Within 12 months, a complete specification is filed detailing the algorithm, system architecture, and specific use cases.

Step 4: Six months later, the team publishes a paper at an IEEE conference (covered by grace period).

Step 5: Examination request is filed within 48 months, and the FER cites prior art in cryptographic protocols.

Step 6: The team responds by amending claims to focus on the specific technical advancement (reduced verification time) supported by experimental data, demonstrating an inventive step.

Step 7: Patent is granted, and the university technology transfer office initiates licensing discussions with fintech companies and blockchain consortia.

Step 8: Concurrently, a PCT application is filed within 12 months of the priority date, with national phase entries planned in the United States, EPO, Japan, and China.

This example illustrates the integration of technical innovation, procedural rigor, and strategic commercialization.

12. Summary

This lecture has traversed the complete patenting lifecycle from patentability criteria through to commercialization. The key takeaways are:

1. **Patentability rests on three pillars:** novelty, inventive step, and industrial application, with statutory exclusions (particularly Section 3(k) for computer programs per se in India) shaping eligibility.
2. **Filing strategy** typically begins with a provisional specification to secure priority, followed by a complete specification within 12 months containing full disclosure and claims.
3. **Examination is request-based** in India, with substantive review by examiners requiring response to objections within prescribed timelines.
4. **Publication at 18 months** triggers provisional rights and opens pre-grant opposition windows.
5. **Opposition mechanisms** (both pre-grant and post-grant) enable third parties to challenge patentability, requiring applicants to maintain robust prosecution records.
6. **Grant confers exclusive rights** for a 20-year term, contingent on renewal fee payment.

7. **Commercialization** through licensing, assignment, or direct exploitation transforms legal rights into economic value, requiring careful valuation and strategic positioning.
8. **International filing** via PCT or Paris Convention extends protection across jurisdictions, with strategic decisions driven by market potential and competitive landscape.

For computer science researchers, mastering this process is essential not only for protecting individual innovations but also for contributing to institutional IP portfolios that support technology transfer, industry partnerships, and national innovation ecosystems.

13. Exam Preparation Tips and Practice Questions

13.1 Preparation Strategies

1. **Create comparative tables** distinguishing provisional vs. complete specifications, pre-grant vs. post-grant oppositions, and different licensing arrangements.
2. **Trace the procedural timeline** on paper to internalize the chronological sequence of filing, publication, examination, and grant.
3. **Practice application-based questions** involving CS inventions to test your ability to apply patentability criteria.
4. **Refer to primary legislation** (Indian Patents Act, 1970 with 2005 amendments) and relevant sections for citation accuracy.
5. **Stay updated** on recent case law, particularly from the Delhi and Madras High Courts' IP Divisions, which handle major patent disputes.

13.2 Key Sections to Remember

Section	Subject
Section 2(1)(j)	Definition of "new invention"
Section 2(1)(ja)	Definition of "inventive step"
Section 3	Inventions not patentable
Section 9	Provisional specification
Section 10	Complete specification contents
Section 11A	Publication of application
Section 11B	Request for examination
Section 25(1)	Pre-grant opposition
Section 25(2)	Post-grant opposition
Section 43	Grant of patent
Section 48	Rights of patentee
Section 53	Term of patent

13.3 Practice Questions

Short Answer Questions:

1. Differentiate between a provisional specification and a complete specification. Under what circumstances is a provisional specification preferred?
2. Explain the concept of "inventive step" with reference to Section 2(1)(ja) of the Indian Patents Act.
3. Why is software classified as non-patentable under Section 3(k) of the Indian Patents Act, and what exceptions allow certain software-related inventions to be patented?
4. Describe the role of the First Examination Report (FER) in patent prosecution.

Long Answer Questions:

1. Critically analyze the procedural steps involved in obtaining a patent in India, from the filing of the application to the grant of the patent. Highlight the role of the Patent Cooperation Treaty (PCT) in international patent filing strategy.

2. Discuss the various grounds for pre-grant and post-grant opposition under the Indian Patents Act. How do opposition proceedings impact the patenting strategy of a technology company?
3. Examine the commercialization pathways for patented computer science inventions. What are the strategic considerations in choosing licensing versus assignment versus direct commercialization?
4. Analyze the patentability criteria under the Indian Patents Act with specific reference to the challenges faced by software and algorithm-based inventions in meeting these criteria.

Application-Based Questions:

1. A group of students develops a novel machine learning algorithm that significantly reduces training time for deep neural networks on edge devices. They wish to patent it in India and file internationally. Outline a step-by-step patenting strategy, including documentation requirements, specification drafting considerations, and commercialization options.
2. After filing a patent application for a distributed consensus protocol, a competitor files a pre-grant opposition citing prior art from an arXiv preprint by a different research group. Analyze the likely grounds for opposition, the procedural implications, and the strategies the applicant may adopt to defend the patent.
3. A university spin-off company wishes to commercialize a patented cybersecurity solution. Compare the relative advantages of licensing, assignment, and spin-off commercialization models for this case, recommending the most suitable path with justification.

13.4 Additional Reading

- Indian Patents Act, 1970 (as amended up to 2005 and subsequent rules)
- WIPO PCT Applicant's Guide
- Manual of Patent Office Practice and Procedure (IPO India)
- P. Narayanan, Intellectual Property Law (Eastern Law House)
- V.K. Ahuja, Law Relating to Intellectual Property Rights (LexisNexis)

End of Lecture Notes

These notes have been prepared for academic instruction under UPES University guidelines and are intended to support the learning objectives of CSEG3060_3 Research Methodology in Computer Science. They should be read in conjunction with prescribed textbooks, statutory texts, and contemporary case law.